

DAS Partner Brief

Dependable Agentic Systems at the University of Delaware

Department of Computer & Information Sciences

Build agents that leave an audit trail

Scientific and operational work needs agentic systems that can explain what they did, what they used, and where a human should intervene.

Most agent demos optimize for apparent autonomy. In research and enterprise operations, the costly failure mode is untraceable automation: an agent changes code, cites an unverifiable source, acts on stale local context, or produces a result that cannot be audited. DAS treats agent dependability as a systems engineering problem: provenance, local knowledge, controllable execution, and model choice.

A strong first use case is not “deploy an agent everywhere.” It is one workflow where the inputs, tools, approval points, and success criteria can be made explicit.

How to start

- Pick one high-value workflow where mistakes are expensive but review is possible.
- Make the context explicit: documents, datasets, tools, permissions, and “done” criteria.
- Require provenance by default: sources, code changes, data transformations, and human approvals.
- Keep sensitive local knowledge local; use cloud or local models according to risk.

Dong Dai — Associate Professor, Co-Director of DAS

Representative work: IOAgent (IPDPS’25), STELLAR (SC’25), ION (HotStorage’24)

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Start from a scientific workflow, not a generic chatbot

Agents add value when they combine domain context, structured data, and human-in-the-loop judgment.

For science and health, the agent is rarely the final decision maker. The useful role is coordination: retrieving the right local evidence, preparing analyses, tracking assumptions, surfacing anomalies, and helping experts move from raw data to testable hypotheses. That requires domain-aware workflows, not a general-purpose chat interface placed on top of messy data.

DAS is building partnerships where agent systems can support domain scientists while preserving expert control. The goal is to make scientific work more reproducible, auditable, and easier to scale across students, labs, and collaborators.

How to start

- Define success in domain terms: better triage, faster analysis, clearer provenance, or fewer manual handoffs.
- Identify the local knowledge the agent must respect: protocols, papers, datasets, code, and lab conventions.
- Keep the expert in the loop at points where interpretation or risk matters.
- Validate outputs against domain baselines before expanding the workflow.

Yixiang Deng — Assistant Professor, Co-Director of DAS

Representative venues: Nature Communications, Cell Reports, npj Digital Medicine

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Assume every agent needs a threat model

The more useful an agent becomes, the more authority and attack surface it inherits.

A dependable agent is not only a better prompt or a stronger model. It is a software system connected to files, APIs, credentials, databases, instruments, and human workflows. Each connection creates a trust boundary. Agent protocols and tool ecosystems, including emerging interfaces such as MCP, make this boundary easier to cross and easier to misunderstand.

DAS security work asks what can go wrong before an agent is placed inside a real organization. That includes adversarial prompts, poisoned tools or documents, overbroad permissions, insecure execution, supply-chain exposure, and audit gaps.

How to start

- Map the tools, permissions, data flows, and approval points before automation.
- Separate low-risk assistance from high-authority actions.
- Log prompt, data, tool, and execution traces for later review.
- Test the workflow against misuse and failure cases before deployment.

Xing Gao — Assistant Professor, DAS Faculty Member

Representative work: MCP security (DSN'26), GPU cache attacks (USENIX Security'24), CI/CD security (CCS'24 / IEEE S&P'24)

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